

## Keys

1. Which of the following does not use symmetric keys? ?
  - A. SEAL
  - B. DES
  - C. RC6
  - D. RSA**
2. What kind of RSA key typically decrypts data in a file transfer? ?
  - A. Symmetric
  - B. Public
  - C. Private**
  - D. Secret
3. Which of the following is not true of asymmetric keying algorithms? ?
  - A. The private key can be shared without compromising security.**
  - B. They are more complex than symmetric algorithms.
  - C. They are more secure than symmetric algorithms.
  - D. They support longer key lengths than symmetric algorithms.
4. Which of the following is not an example of an asymmetric keying algorithm? ?
  - A. DSS
  - B. Skipjack**
  - C. DH
  - D. ECC

## Answers

1.
  - ☒ **D.** RSA uses asymmetric keys.
  - ☒ **A, B,** and **C** use symmetric keys.
2.
  - ☒ **C.** Private keys decrypt data.
  - ☒ RSA uses asymmetric keys, not symmetric, making **A** incorrect. **B** encrypts data. **D** is a nonexistent symmetric/asymmetric keying term.
3.
  - ☒ **A.** The public, not private, key can be shared without compromising security.
  - ☒ **B, C,** and **D** are true of asymmetric keys and their algorithms.
4.
  - ☒ **B.** Skipjack uses a symmetric algorithm.
  - ☒ **A, C,** and **D** are examples of asymmetric algorithms.

## Ciphers

5. Which of the following is not true of 3DES? ?
  - A. It encrypts blocks of 64-bit data.
  - B. It uses ECB and CBC modes.
  - C. It runs DES three times.
  - D. It uses a substitution cipher.**
6. Which of the following ciphers operates on small chunks of data, typically in bit sizes? ?
  - A. Block

- B. Stream**
- C. Transposition
- D. Polyalphabetic

## Answers

- 5.
  - ☒ **D.** 3DES uses a block, not a substitution, cipher.
  - ☒ **A, B,** and **C** are true of 3DES.
- 6.
  - ☒ **B.** A stream cipher operates on small chunks of data, typically in bit sizes.
  - ☒ **A** operates on blocks of data. **C** encrypts data by taking a unit of plaintext characters, commonly called a group, and shifts them in a regular pattern that results in a permutation. **D** performs substitution using multiple substitution alphabets.

## Technology Implementation

- 7. A(n) \_\_\_\_\_ is where small changes in the input data create large changes in the resulting output data. ?
  - A. Asymmetric key
  - B. Avalanche effect**
  - C. Electronic code book
  - D. Cipher-block chaining
- 8. What takes variable-length input, along with a key, and creates a fixed-length output? ?
  - A. Encryption algorithm
  - B. HMAC function** {Hash Message Authentication Code}
  - C. IDEA
  - D. RC4
- 9. How long are MD5 signatures? ?
  - A. 64 bits
  - B. 112 bits
  - C. 128 bits**
  - D. 160 bits
- 10. Which of the following is an example of an HMAC function? ?
  - A. Skipjack
  - B. RC2
  - C. DH
  - D. SHA-1**

## Answers

- 7.
  - ☒ **B.** An avalanche effect is where small changes in the input data create large changes in the resulting output data.
  - ☒ **A** is used to encrypt and decrypt data or create and verify digital signatures. **C** divides data into fixed-length blocks, where each block is encrypted separately. **D** takes each block of data and XORs it with the

previous ciphertext block before encrypting it.

- 8.
  - ☒ **B**. An HMAC function takes variable-length input, along with a key, and creates a fixed-length output.
  - ☐ **A**, **C**, and **D** take input and produce the same length output.
- 9.
  - ☒ **C**. MD5 signatures are 128 bits in length.
  - ☐ Since MD5 signatures are 128 bits in length, answers **A**, **B**, and **D** are incorrect.
- 10.
  - ☒ **D**. SHA-1 and MD5 are examples of HMAC functions.
  - ☐ **A**, **B**, and **C** are examples of encryption algorithms.

Date: July 8 , 2013

**Module Name: PKI & Biometric**

- Q1. Find the pair of commands which will produce same message digest of a file in Linux machine. ?  
a) md4sum , md5sum  
b) sha1sum , sha160sum  
**c) shasum , sha1sum**  
d) sha160sum , shasum
- Q2. Which of the following algorithm used in SSL /TLS Protocol. ?  
a) AES &DES both  
b) (b)RSA only  
c) (c)DES & RSA both  
**d) (D) AES & RSA both**
- Q3. How many mixing and mashing rounds will be there in RC2 Algorithm?  
a) 18, 2  
**b) 16, 2**  
c) 16, 4  
d) 2, 15
- Q4. How many servers are used in Kerberos for authentication purpose?  
a) 4  
**b) 3**  
c) 1  
d) 2
- Q5. Diffie-Hellman algorithm's effectiveness depends on the difficulty of computing?  
**a) Discrete logarithms**  
b) Prime factors of composite number  
c) Elliptic curve  
d) Abelian group
- Q6. The total Key Size used in RSA algorithm is.....?  
a) 128 bits  
b) 256 bits  
c) 512 bits  
**d) 1024 bits**
- Q7. Which of the following statements are true about OCSP and CRL?  
(I) The CRL is a list of subscribers paired with digital certificate status.  
(II) The main limitation of CRL is the fact that updates must be frequently downloaded to keep the list current.  
(III) The OCSP checks certificate status in real time.  
(IV) The CRL allows the authenticity of a certificate to be immediately verified.  
  
a) i only  
b) both i and ii  
c) i, ii, iv only  
**d) i, ii, iii only**
- Q8. To exchange e-mail messages, using PGP a user needs of .....?  
a) secret key  
**b) Public key**  
c) both (a) & (b)  
d) none
- Q9. Which of the following protocol is used security and compression services to data generated from the application layer. ?  
**a) SSL/TLS**  
b) IPSec  
c) PGP  
d) none
- Q10. What technology is being used to detect anomalies?  
a) CVE  
b) Sniffing  
**c) IDS**  
d) Capturing
- Q11. Why would a digital certificate be added to a certificate revocation list (CRL)?  
a) If the public key had become compromised  
**b) If the private key had become compromised**  
c) The certificate had become public  
d) None of the above.  
e)
- Q12. What does OCSP stand for with respect to digital certificates?  
**a) Online Certificate Status Protocol**  
b) Online Certificate Service Provider  
c) Open Certificate Systems Project  
d) Open Certificates for Security and Privacy
- Q13. Which of these is not an encryption algorithm?  
**a) Diffie hellman**  
b) AES  
c) DES  
d) RSA
- Q14. In which mode of operation, IPSec protects information delivered from the transport layer to the network layer. ?  
a) Transport  
b) (b)tunnel  
c) (c) both a & b  
d) (d) none
- Q15. Which of the following provides authentication at IP layer of OSI?  
a) AH  
b) ESP  
**c) AH and ESP both**  
d) SSL
- Q16. Which of the following correct about SSL

Date: July 8 , 2013

- Protocol, it provides
- a) message integrity
  - b) Compression
  - c) Confidentiality
  - d) All of the above**
- Q17. In e-mail system which of one security protocol is used for sending and receiving message. ?
- a) SSL
  - b) IPSec
  - c) PGP**
  - d) SSL and PGP both
- Q18. Which of the following protocol provides either authentication or encryption, or both, for packets at the IP level?
- a) SSL
  - b) ESP**
  - c) AH
  - d) PGP
- Q19. The combination of key exchange, hash, and encryption algorithms defines a \_\_\_\_\_ for each SSL session. ?
- a) A suite of protocol
  - b) List of protocol
  - c) Cipher suite**
  - d) Combination of keys
- Q20. Kerberos is an ----- protocol?
- a) Session initiation
  - b) Authentication**
  - c) Key generation
  - d) None of the above
- Q21. In which way does the Combined Encryption combine symmetric and asymmetric encryption?
- a) First, the message is encrypted with symmetric encryption and afterwards it is encrypted asymmetrically together with the key.**
  - b) The secret key is symmetrically transmitted, the message itself asymmetrically.
  - c) First, the message is encrypted with asymmetric encryption and afterwards it is encrypted symmetrically together with the key
  - d) The secret key is asymmetrically transmitted, the message itself symmetrically.
- Q22. Which is the largest disadvantage of the symmetric Encryption?
- a) More complex and therefore more time-consuming calculations
  - b) Problem of the secure transmission of the Secret Key.
  - c) Less secure encryption function.**
  - d) Isn't used any more
- Q23. Mishra is a certified ethical hacker working

in CDAC, one day Mishra finds one employee that appears to be sending very large email to some other marketing company that recently underwent a string of thefts and corporate espionage incidents, even though they should have no reason to be communicating with them. Tracks down the actual emails sent and upon opening them, only finds picture files attached to them. These files seem perfectly harmless, usually containing some kind of joke. Mishra decides to use some special software to further examine the pictures and finds that each one had hidden text that was stored in each picture

What technique was used by the CDAC employee to send information to the rival marketing company?

- a) The CDAC employee used cryptography to hide the information in the emails sent
  - b) The method used by the employee to hide the information was logical watermarking
  - c) The employee used steganography to hide information in the picture attachments**
  - d) By using the pictures to hide information, the employee utilized picture fuzzing
- Q24. Mayank work for CDAC as a Sales Manager. The company has tight network security restrictions. Mayank is trying to steal data from the company's Sales database (Sales.xls) and transfer them to his home computer. Mayank's company filters and monitors traffic that leaves from the internal network to the Internet. How will Mayank achieve this without raising suspicion?
- a) Encrypt the Sales.xls using PGP and e-mail it to your personal gmail account
  - b) You can conceal the Sales.xls database in another file like photo.jpg or other files and send it out in an innocent looking email or file transfer using Steganography techniques.**
  - c) Package the Sales.xls using Trojan wrappers and telnet them back your home computer
  - d) Change the extension of Sales.xls to sales.txt and upload them as attachment to your hotmail account

Q25. SSL has been seen as the solution to

Date: July 8 , 2013

a lot of common security problems. Administrator will often time make use of SSL to encrypt communications from points A to point B. Why do you think this could be a bad idea if there is an Intrusion Detection System deployed to monitor the traffic between point A and B?

- a) SSL is redundant if you already have IDS's in place
- b) SSL will trigger rules at regular interval and force the administrator to turn them off
- c) SSL will slow down the IDS while it is breaking the encryption to see the packet content
- d) SSL will blind the content of the packet and Intrusion Detection Systems will not be able to detect them**

Q26. \_\_\_\_\_ is a type of symmetric-key encryption algorithm that transforms a fixed-length block of plaintext (unencrypted text) data into a block of cipher text (encrypted text) data of the same length.

- a) Stream Cipher
- b) Hash Cipher
- c) Block Cipher**
- d) Bit Cipher

Q27. One of the most common and the best way of cracking RSA encryption is to begin to derive the two prime numbers, which are used in the RSA PKI mathematical process. If the two numbers p and q are discovered through a \_\_\_\_\_ process, then the private key can be derived.

- a) Factorization**
- b) Brute-forcing
- c) Prime Detection
- d) Hashing

Q28. In SSL, in which of the following choices does authentication happen through the Exchange of certificates?

- a) Server
- b) Client
- c) Both A and B**
- d) Neither A nor B

Q29. Which component of a firewall provides filtering based on the IP addresses and types of connections (transport layer) information stored in the packet?

- a) Packet filtering
- b) Application proxies**
- c) Circuit level gateways
- d) Virtual private networking

Q30. The \_\_\_\_\_ attack can endanger the

security of the Diffie-Hellman method if two parties are not authenticated to each other.

- a) man-in-the-middle**
- b) cipher text attack
- c) plaintext attack
- d) Ping flood

Q31. A digital signature implementation needs a \_\_\_\_\_ system.

- a. symmetric-key
- b. asymmetric-key**
- c. (a) or (b)
- d. Both (a) and (b)

Q32. Using public-key cryptography, suppose Raghav wants to send a message to Neelima, and Neelima wants to be sure that the message was indeed sent by Raghav only. Then Raghav should (choose the appropriate scenario)

- a) Encrypt the message with Neelima's private key and send the encrypted message to Neelima.
- b) Encrypt the message with Neelima's public key and send Neelima the message.
- c) Encrypt the message with his public key and send Neelima the message.
- d) Encrypt the message with his private key and send the encrypted message to Neelima.**

Q33. When you receive a public key that has been signed by a number of individuals, that key is part of ...?

- a) The web of trust.**
- b) a certificate authority
- c) a digital fingerprint.
- d) an illegal scam

Q34. An \_\_\_\_\_ organization known as \_\_\_\_\_ sends out information about known security holes in software

- a) RSA
- b) PKI
- c) CERT**
- d) PGP

Q35. A firewall is .....?

- a) An established network performance reference point.
- b) Software or hardware used to isolate a private network from a public network.**
- c) A virus that infects macros.
- d) A predefined encryption key used to encrypt and decrypt data transmissions

Q36. Firewalls operate by .....?

- a) The pre-purchase phase.
- b) Isolating Intranet from Extranet.
- c) Screening packets to/from the Network and provide controllable filtering of network traffic.**
- d) None of the above.

Q37. Mechanism to protect private networks

Date: July 8 , 2013

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from outside attack is

- a) **Firewall**
- b) Antivirus
- c) Digital signature
- d) Formatting

Q38. A protocol developed by Netscape which handles data encryption in a transparent manner within the Web browser is called....

- a) Netscape Navigator.
- b) **Secure socket layer.**
- c) IPSec.
- d) VeriSign.

Q39. What is the port number of SSL Protocol?

- a) 433
- b) 469
- c) **443**
- d) 25

Q40. PKCS stand for ...?

- a) Private-Key Cryptography Standard
- b) **Public-Key Cryptography Standard**
- c) Public-Key Cryptography Scheme
- d) Both (a) and (b)

## ***Cryptography Questions***

Q1 In substitution cipher ,

- (a) characters are replaced by other characters**
- (b) columns are replaced by rows
- (c) rows are replaced by columns
- (d) none of the above

Q2. Q6 Redundancy of plain text increases in

- (a) diffusion**
- (b) confusion
- (c) both confusion & diffusion
- (d) neither confusion nor diffusion

Q3 . If  $p=11$  and  $q=17$ , the public and private keys for encrypting through RSA algorithm are

- (a) (7,187) and (23,87)
- (b) (3,187) and (107,187)
- (c) (5,187) and (23, 187)
- (d) Both a and b**

Q4 Vernam cipher is also called as

- (a) rail fence technique
- (b) book cipher
- (c) one time pad**
- (d) running key cipher
- (e)

Q5 The no. of unique keys that could be generated with a 3 bit key is

- (a) 6
- (b) 8**
- (c) 9
- (d) 12

Q6. Q17 Playfair Cipher is an example of

- (a) A Substitution cipher**
- (b) Transposition cipher
- (c) Both of above
- (d) None of above

Q7. Knowledge based authentication is based on

- (a) what the users possess
- (b) what the users know**
- (c) what the users are
- (d) all of the above

Q8 Which technique is used for hiding the message

- (a) Cryptography
- (b) Steganography**
- (c) Cryptology
- (d) Both a and b

Q9. The concept that is crucial for asymmetric cryptography is

- (a) integers
- (b) prime numbers**
- (c) negative numbers
- (d) fractions

Q10. The three primary attributes of information security are

- (a) confidentiality, integrity & availability**
- (b) authentication, authorization & non-repudiation
- (c) confidentiality, access control & non-repudiation
- (d) availability, access control & authorization

Q11 Infectors reside in specific areas of the PC's hard disk, those that are read and executed by the computer at boot time

- a) Boot Sector Viruses**
- b) Resident viruses
- c) Non-resident viruses
- d) File Infectors.

Q12. Which technique is used for hiding the message

- a) Cryptography
- b) Steganography**
- c) Cryptology
- d) Both a and b

Q13. Malicious software that spreads across networks

- a) Virus
- b) Worm**
- c) Backdoor
- d) Trapdoor

Q14. Which of the following is not a biometric I & A technique?

- a. Password**
- b. Fingerprint
- c. Voice
- d. Retina scan

Q15. Vernam Cipher is the example of

- (a) Substitution cipher**
- (b) Transposition cipher
- (a) Both of above
- (b) None of above

Q16. Complex file infectors that change physical forms

**A Polymorphic virus**

B Boot Sector Virus

C Stealth Virus

D Non-resident Virus

Q17 Firewall should be situated

- (a) inside a corporate network
- (b) outside a corporate network
- (c) between a corporate network and the outside world**
- (d) none of these

Q18. Cryptanalyst is a person who

- (a) devises cryptography solutions
- (b) attempts to break cryptography solutions**
- (c) none of these
- (d) both of these

Q19. Allowing certain users specific accesses implies implementation of

- (a) confidentiality
- (b) integrity
- (c) authentication
- (d) access control**

Q20. What does a hash function do

- a) Creates a secure tunnel
- b) Breaks encryption by trying every possible key
- c) Multiplies two very large primes
- d) **Creates a unique digest of a message**

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**Date:**

**Module Name: PKI**

Q. No. 1

**Question:**

Atbash is an example of a \_\_\_\_\_

Answer Choices

**A: Substitution Cipher**

B: Transposition Cipher

C: All of the above

D: None of the above

Q. No. 2

**Question:**

\_\_\_\_\_ cipher replaces bits, characters, or blocks of characters with different bits, characters, or blocks.

Answer Choices

**A: Substitution**

B: XORing

C: Transposition

D: Transportation

Q. No. 3

**Question:**

Father of Modern Cryptography

Answer Choices

A: Archie Walls

B: William Frederick Friedman

C: John Warren

D: Donald N Wilber

Q. No. 4

**Question:**

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\_\_\_\_\_ refers to the set of all possible keys that can be used to initialize it.

Answer Choices

- A: Cryptology
- B: Cryptanalysis
- C: Key Space
- D: Encryption algorithm

Q. No. 5

**Question:**

20-bit key would have a key space of \_\_\_\_\_

Answer Choices

- A: 1084572
- B: 1048578
- C: 1084576
- D: 1048576

Q. No. 6

**Question:**

The frequency of brute force attacks has increased because:

Answer Choices

- A: The use of permutations and transpositions in algorithms has increased.
- B: As algorithms get stronger, they get less complex, and thus more susceptible to attacks.
- C: Processor speed and power has increased.
- D: Key length reduces over time.

Q. No. 7

**Question:**

What is the primary purpose of using one-way hashing on user passwords?

Answer Choices

- A: It minimizes the amount of primary and secondary storage needed to store passwords.

- 
- B: It prevents anyone from reading passwords in plaintext.  
C: It avoids excessive processing required by an asymmetric algorithm.  
D: It prevents replay attacks.

Q. No. 8

**Question:**

What is the definition of an algorithm's work factor?

Answer Choices

- A: The time it takes to encrypt and decrypt the same plaintext.  
B: The time it takes to break the encryption  
C: The time it takes to implement 16 rounds of computation.  
D: The time it takes to apply substitution functions

Q. No. 9

**Question:**

Which of the following is a U.S. federal government algorithm developed for creating secure message digests?

Answer Choices

- A: Data Encryption Algorithm  
B: Digital Signature Standard  
C: Secure Hash Algorithm  
D: Data Signature Algorithm

Q. No. 10

**Question:**

What does DEA stand for?

Answer Choices

- A: Data Encoding Algorithm  
B: Data Encoding Application  
C: Data Encryption Algorithm  
D: Digital Encryption Algorithm

Q. No. 11

**Question:**

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If different keys generate the same ciphertext for the same message, what is this called?

Answer Choices

- A: Collision
- B: Secure hashing
- C: MAC
- D: Key clustering

Q. No. 12

**Question:**

What does DES stand for?

Answer Choices

- A: Data Encryption System
- B: Data Encryption Standard
- C: Data Encoding Standard
- D: Data Encryption Signature

Q. No. 13

**Question:**

Many countries restrict the use or exportation of cryptographic systems. What is the reason given when these types of restrictions are put into place?

Answer Choices

- A: Without standards, there would be many interoperability issues when trying to employ different algorithm in different programs.
- B: The systems can be used by some countries against their local people.
- C: Criminals could use encryption to avoid detection and prosecution.
- D: Laws are way behind, so adding different types of encryption would confuse the laws more.

Q. No. 14

**Question:**

Which of the following is a true statement pertaining to data encryption when it is used to protect data?

Answer Choices

- A: It verifies the integrity and accuracy of the data.

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B: It requires careful key management.

C: It does not require much system overhead in resources.

D: It requires keys to be escrowed.

Q. No. 15

**Question:**

Which of the following is not a property or characteristic of a one-way hash function?

Answer Choices

A: It converts a message of arbitrary length into a value of fixed length.

B: Given the digest value, it should be computationally infeasible to find the corresponding message.

C: It should be impossible or rare to derive the same digest from two different messages.

D: It converts a message of fixed length to an arbitrary length value.

Q. No. 16

**Question:**

The generation of keys that are made up of random values is referred to as Key Derivation Function (KDFs). What values are not commonly used in this key generation process?

Answer Choices

A: Hashing values

B: Asymmetric values

C: Salts

D: Passwords

Q. No. 17

**Question:**

What is the goal of cryptanalysis?

Answer Choices

A: To determine the strength of an algorithm

B: To increase the substitution function in a cryptographic algorithm.

C: To decrease the transposition function in a cryptographic algorithm.

D: To determine the permutations used.

Q. No. 18

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**Question:**

How many bits make up the effective length of the DES key?

Answer Choices

A: 56

B: 64

C: 32

D: 16

Q. No. 19

**Question:**

Who was involved in developing the first public key algorithm?

Answer Choices

A: Adi Shamir

B: Ross Anderson

C: Bruce Schneier

D: Martin Hellman

Q. No. 20

**Question:**

Which of the following uses a symmetric key and a hashing algorithm?

Answer Choices

A: HMAC

B: Triple-DES

C: ISAKMP-OAKLEY

D: RSA

Q. No. 21

**Question:**

The word Cryptography is derived from the Greek word \_\_\_\_\_, meaning "hidden, secret".

Answer Choices

A: Krios

B: Kyros

C: Kerberos

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D: Kryptos

Q. No. 22

**Question:**

DES performs how many rounds of permutation and substitution?

Answer Choices

A: 16

B: 32

C: 64

D: 56

Q. No. 23

**Question:**

Kerckhoffs principle states that \_\_\_\_\_

Answer Choices

A: the keyspace should be public

B: the algorithm should be public

C: the key should be public

D: none of the above

Q. No. 24

**Question:**

In symmetric algorithms, if 100 people need to communicate securely with each other, \_\_\_\_\_ keys are required

Answer Choices

A: 3820

B: 4480

C: 4950

D: 3480

$100 \cdot (100-1)/2$

Q. No. 25

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**Question:**

Which of the following best describes a certificate authority?

Answer Choices

A: An organization that issues private keys and the corresponding algorithms

B: An organization that validates encryption processes.

C: An organization that verifies encryption keys.

D: An organization that issues certificates

Q. No. 26

**Question:**

Simple substitution and transposition ciphers are vulnerable to attacks that perform

\_\_\_\_\_

Answer Choices

A: Ping of Death

B: Frequency Analysis

C: Denial of Service Attack

D: SQL Injection

Q. No. 27

**Question:**

Which of the following is based on the fact that it is hard to factor large numbers into two original prime numbers?

Answer Choices

A: ECC

B: RSA

C: DES

D: Diffie-Hellman

Q. No. 28

**Question:**

\_\_\_\_\_ is mono-alphabetic substitution while \_\_\_\_\_ is not

Answer Choices

A: Caesar, Vigenere

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B: Vigenere , Caesar

C: MD4, MD5

D: MD5, MD4

Q. No. 29

**Question:**

What would indicate that a message had been modified?

Answer Choices

A: The public key has been altered.

B: The private key has been altered.

C: The message digest has been altered.

D: The message has been encrypted properly.

Q. No. 30

**Question:**

Which of the following describes the difference between the Data Encryption Standard and the Rivest-Shamir-Adleman algorithm?

Answer Choices

A: DES is symmetric, while RSA is asymmetric.

B: DES is asymmetric, while RSA is symmetric.

C: They are hashing algorithms, but RSA produces a 160-bit hashing value.

D: DES creates public and private keys, while RSA encrypts messages.

Q. No. 31

**Question:**

In a Poly-alphabetic substitution cipher, \_\_\_\_\_ alphabets are used to substitute another.

Answer Choices

A: One

B: Multiple

C: All of the above

D: None of the above

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Q. No. 32

**Question:**

Which of the following best describes a digital signature?

Answer Choices

A: A method of transferring a handwritten signature to an electronic document.

B: A method to encrypt confidential information.

C: A method to provide an electronic signature and encryption.

D: A method to let the receiver of the message prove the source and integrity of a message.

Q. No. 33

**Question:**

\_\_\_\_\_ cipher does not replace the original text with different text, but rather moves the original values around. It re-arranges the bits, characters, or blocks of characters to hide the original meaning.

Answer Choices

A: Substitution

B: XORing

C: Transposition

D: Transportation

Q. No. 34

**Question:**

What is used to create a digital signature?

Answer Choices

A: The receiver's private key.

B: The sender's public key.

C: The sender's private key.

D: The receiver's public key.

Kommentar [1]:

Q. No. 35

**Question:**

Which of the following best describes the difference between HMAC and CBC-MAC?

Answer Choices

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A: HMAC creates a message digest and is used for integrity; CBC-MNAC is used to encrypt blocks of data for confidentiality.  
B: HMAC uses a symmetric key and a hashing algorithm; CBC-MAC uses the first block for the checksum.  
C: HMAC provides integrity and data origin authentication; CBC-MAC uses a block cipher for the process of creating a MAC.  
D: HMAC encrypts a message with a symmetric key and then puts the result through a hashing algorithm; CBC-MAC encrypts the whole message.

Q. No. 36

**Question:**

Why would a certificate authority revoke a certificate?

Answer Choices

- A: If the user's public key has become compromised
- B: If the user changed over to using the PEM model that uses a web of trust.
- C: If the user's private key has become compromised.
- D: If the user moved to a new location

Q. No. 37

**Question:**

Another name for cryptography strength.

Answer Choices

- A: Packet length
- B: Cryptographic Key
- C: Key Exchange Factor
- D: Work Factor

Q. No. 38

**Question:**

\_\_\_\_\_ is known as Caesar Cipher on Steroids.

Answer Choices

- A: Hieroglyphs
- B: Atbash
- C: Vigenere
- D: Scytale

---

Q. No. 39

**Question:**

Science of studying and breaking the secrecy of encryption processes, compromising authentication schemes, and reverse-engineering algorithms and keys.

Answer Choices

- A: Cryptology
- B: Cryptanalysis
- C: Encryption
- D: Cryptosystem

Q. No. 40

**Question:**

What is an advantage of RSA over the DSA?

Answer Choices

- A: It can provide digital signature and encryption functionality.
- B: It uses fewer resources and encrypts faster because it uses symmetric keys.
- C: It is a block cipher rather than a stream cipher.
- D: It employs a one-time encryption pad.

## Site-to-Site Preparation

1. Enter the protocol and port that must be specified within an interface ACL to allow a management connection to be terminated on the router: ?
- \_\_\_\_\_.

## Answers

1.
  - ☒ The protocol and port that must be specified within an interface ACL to allow a management connection to be terminated on the router is UDP 500.

## ISAKMP/IKE Phase 1 Configuration

2. If you omit a parameter in an ISAKMP/IKE Phase 1 policy, a default value is used. Which of the following is not a default value? ?
- A. MD5  
B. DES  
C. DH group 1  
D. RSA signatures
3. You need to create an ISAKMP/IKE Phase 1 policy that uses the following parameters: preshared keys, using a preshared key of abc123 for peer 192.1.1.1, 3DES, MD5, DH key group 2, and a lifetime of one hour. Use a policy number of 100. Enter the commands to accomplish this: ?
- \_\_\_\_\_.

## Answers

2.
  - ☒ A. SHA-1 is the default, not MD5, if the HMAC function is omitted from a Phase 1 policy.
  - ☒ Answers B, C, and D are default values, if omitted.
3.
  - ☒ Here is the configuration to set up the policy:
  - 
  - Router(config)# **crypto isakmp policy 100**
  - Router(config-isakmp)# **authentication pre-share**
  - Router(config-isakmp)# **encryption 3des**
  - Router(config-isakmp)# **group 2**
  - Router(config-isakmp)# **hash md5 sha**
  - Router(config-isakmp)# **lifetime 3600**
  - Router(config-isakmp)# **exit**
  - Router(config)# **crypto isakmp key 0 abc123 address 192.1.1.1**

## ISAKMP/IKE Phase 2 Configuration

4. What is the default lifetime of data SAs, if not overridden? ?
- A. 300 seconds

- B. 600 seconds
  - C. 3,600 seconds**
  - D. 86,400 seconds
5. What is the default mode for a transform set, if omitted? ?
- A. Transport
  - B. Transform
  - C. Symmetrical
  - D. Tunnel**
6. Which of the following is not required in a crypto map entry? ?
- A. Crypto ACL
  - B. Router's local IP address**
  - C. Transform set name
  - D. Connection method: manual or dynamic (ISAKMP/IKE)
7. Which of the following traffic is dropped when using static crypto maps and IPSec? ?
- A. Inbound traffic that is protected and matches a crypto ACL entry
  - B. Inbound traffic that doesn't match a crypto ACL entry
  - C. Inbound traffic that matches a crypto ACL entry and is in clear text**
  - D. Outbound traffic that matches a crypto ACL and is already protected by IPSec

## Answers

4. ☒ **C.** The default lifetime of data SAs, if not overridden, is 3,600 seconds.
- ☒ **A** and **B** are too small. **D** is the default value for the management connection.
5. ☒ **D.** The default mode for a transform set, if omitted, is tunnel mode.
- ☒ **A** is used for host-to-host connections and is not the default. **B** and **C** are nonexistent transform set modes.
6. ☒ **B.** The remote peer's IP address, not the local router's address, must be specified in a crypto map entry.
- ☒ Answers **A**, **C**, and **D** are required.
7. ☒ **C.** Inbound traffic that matches a crypto ACL entry and is in clear text is dropped by the router.
- ☒ **A** is verified and decrypted by the router. **B** is not processed by IPSec. **D** is protected by IPSec again.

## Examining and Troubleshooting Connections

8. What state will a management connection be in if it is successfully set up? ?
- A. MM\_NO\_STATE
  - B. QM\_IDLE**
  - C. MM\_SA\_SETUP
  - D. MM\_KEY\_AUTH
9. Which output from the `debug crypto isakmp` command indicates a failed policy negotiation? ?

- A. `%CRYPTO-6-IKMP_SA_NOT_OFFERED: Remote peer % 15i responded with attribute [chars] not offered or changed.`
- B. `%CRYPTO-6-IKMP_SA_NOT_AUTH: Cannot accept Quick Mode exchange from %15i if SA is not authenticated!`
- C. `IPSec(validate_transform_proposal): proxy identities not supported; ISAKMP: IPSec policy invalidated proposal`
- D. `IPSEC(validate_proposal): invalid local address IP_address`

## Answers

- 8.
  - ☒ **B.** `QM_IDLE` indicates that the management connection has completed successfully.
  - ☒ **A** indicates that the tunnel has been initialized but nothing has been negotiated. **C** indicates that the IPSec peers have negotiated the ISAKMP/IKE Phase 1 policies. **D** indicates that the management SA has been authenticated.
- 9.
  - ☒ **A.** The `"%CRYPTO-6-IKMP_SA_NOT_OFFERED: Remote peer % 15i responded with attribute [chars] not offered or changed."` message in the output of the `debug crypto isakmp` command indicates a failed policy negotiation.
  - ☒ **B** indicates Phase 1 authentication failure. **C** indicates a mismatched crypto ACL in Phase 2. **D** indicates an invalid remote peer IP address in Phase 1.

## Classic Method Site-to-Site Example

10. What crypto map command is used to specify the crypto ACL to use?

?

- A. `set transform-set`
- B. `match address`
- C. `set address`
- D. `set crypto-acl`

## Answers

- 10.
  - ☒ **B.** The `match address` crypto map command is used to specify the crypto ACL to use.
  - ☒ **A** defines the transform set to use. **C** and **D** are invalid commands.

1. How many keys exist in a public/private key pair?
  - A. 1
  - B. 2
  - C. 3
  - D. 4
2. How many keys are needed for symmetric key encryption?
  - A. 1
  - B. 2
  - C. 3
  - D. 4
3. Which of the following key lengths would be considered uncrackable?
  - A. 512
  - B. 32
  - C. 128
  - D. 64
4. What algorithm outputs a 128-bit message digest regardless of the length of the input?
  - A. SHA
  - B. MD5
  - C. RC4
  - D. RC6
5. What algorithm outputs a 160-bit key with variable-length input?
  - A. SHA
  - B. MD5
  - C. RC4
  - D. RC6
6. Which algorithm is used in the digital signature process?
  - A. RC4
  - B. RC5
  - C. Blowfish
  - D. MD5
7. What is cryptography?
  - A. The study of computer science
  - B. The study of mathematics
  - C. The study of encryption
  - D. The creation of encryption algorithms
8. What is the process of changing the order of some characters in an encryption key?
  - A. Transposition
  - B. Subtraction
  - C. Substitution
  - D. Transrelation

9. Data encrypted with the server's public key can be decrypted with which key?
  - A. The server's public key
  - B. The server's private key
  - C. The client's public key
  - D. The client's private key
10. Which type of encryption is the fastest to use for large amounts of data?
  - A. Symmetric
  - B. Public
  - C. Private
  - D. Asymmetric
11. What is the goal of a known-plain text attack?
  - A. To read the encrypted data
  - B. To gain access to the public key
  - C. To discover the encryption key
  - D. To validate the sender of the data
12. Which cryptographic attack attempts to crack the code by looking for patterns and using statistical analysis?
  - A. Cipher text-only attack
  - B. Chosen-plain text attack
  - C. Chosen-cipher text attack
  - D. Brute-force attack
13. Which factor is of concern when using brute-force attacks against encryption?
  - A. Time
  - B. Knowledge of receiver
  - C. Knowledge of the sender
  - D. The ability to capture data
14. Which program is useful in ensuring the integrity of a file that has been downloaded from the Internet?
  - A. Tripwire
  - B. Norton Internet Security
  - C. Snort
  - D. WinMD5
15. What are some of the common fields in an x.509 certificate?
  - A. Secret Key
  - B. Expiration Date
  - C. Hacker
  - D. Public Key
16. What is the standard format for digital certificates?
  - A. x.500
  - B. x.509
  - C. x.25
  - D. XOR

17. What would the cipher text result be of a value of 1 in plain text and 0 in the secret key after an XOR process?  
A. 1  
B. 0
18. What are two components of a PKI?  
A. User passwords  
B. Digital certificates  
C. Encrypted data  
D. CA
19. What element of the CIA triad ensures that the data sent is the same data received?  
A. Confidentiality  
B. Integrity  
C. Authentication
20. What is the purpose of a hash?  
A. To ensure confidentiality when using a public network such as the Internet  
B. To ensure integrity of a transferred file  
C. To ensure only authorized users are accessing a file  
D. To ensure the data is available to authorized users
21. If you omit a parameter in an ISAKMP/IKE Phase 1 policy, a default value is used. Which of the following is not a default value?  
A. MD5  
B. DES  
C. DH group 1  
D. RSA signatures
22. What is the default lifetime of data SAs, if not overridden?  
A. 300 seconds  
B. 600 seconds  
C. 3,600 seconds  
D. 86,400 seconds
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B. Transform  
C. Symmetrical  
D. Tunnel
24. Which of the following is not required in a crypto map entry?  
A. Crypto ACL  
B. Router's local IP address  
C. Transform set name  
D. Connection method: manual or dynamic (ISAKMP/IKE)
25. Which of the following traffic is dropped when using static crypto maps and IPSec?  
A. Inbound traffic that is protected and matches a crypto ACL entry  
B. Inbound traffic that doesn't match a crypto ACL entry

- C. Inbound traffic that matches a crypto ACL entry and is in clear text
  - D. Outbound traffic that matches a crypto ACL and is already protected by IPSec
- 26 What is an example of a packet-filtering firewall?
  - A. Websense
  - B. Router ACLs
  - C. ASA state table
  - D. Cut-through proxy
- 27 Which is an advantage of a stateful firewall over a packet-filtering firewall?
  - A. Allows traffic for a connection to return through the firewall
  - B. Is more difficult to implement filtering of fragments
  - C. Supports user authentication of connections
  - D. Is complex to configure filtering policies
- 28 What kind of firewall terminates users' connections and establishes new connections to the actual destination?
  - A. Application inspection firewall
  - B. Stateful firewall
  - C. Packet-filtering firewall
  - D. Application gateway firewall
- 29 What information is found in a state table of a stateful filtering firewall? (Choose two.)
  - A. TCP flags
  - B. Protocol numbers or names
  - C. MAC addresses
  - D. FTP commands executed by a user
- 30 What kind of firewall would be used to allow multiple connections securely through the firewall for a protocol, like RTP?
  - A. Application proxy
  - B. Packet filter
  - C. Application inspection firewall
  - D. Stateful
- 31 You have a router with two interfaces: FA0/0 and FA0/1. FA0/0 has networks 10.0.1.0/24, 10.0.2.0/24, and 192.168.1.0/24 associated with it. FA0/1 has networks 10.0.3.0/24, 192.168.2.0/24, and 192.168.3.0/24 associated with it. Users associated with FA0/1 need to connect to servers to FA0/0. In this situation, what addresses should you drop to prevent spoofing attacks?
  - A. Source addresses from 192.168.2.0/24
  - B. Destination addresses from 192.168.1.0/24
  - C. Destination addresses of 192.168.2.0/24
  - D. Source addresses from 10.0.1.0/24
- 32 Which of the following is true of an IDS?
  - A. Uses inline mode
  - B. Implements blocking to stop an attacker
  - C. Uses IP logging to send alerts
  - D. Only supports signatures to look for an attack

- 33 What IPS/IDS implementation uses templates and rules to find attacks?
- A. Profiles
  - B. Signatures
  - C. Protocol analysis
  - D. Policies
- 34 What is not an action an IDS or IPS can typically take when an attack is detected?
- A. TCP reset
  - B. IP logging
  - C. Produce an alert
  - D. Rate-limit traffic
- 35 Which of the following is not a purpose of a security policy?
- A. Define a baseline.
  - B. Define how to handle security incidents.
  - C. Define what is and is not allowed.
  - D. Define the organizational chart.
- 36 Which of the following security policy components defines what is and is not allowed regarding an organization's assets?
- A. Statement of authority
  - B. Acceptable use
  - C. Identification
  - D. Incident handling
- 37 Which of the following is the most important factor in a security design?
- A. Risk analysis
  - B. Security policy
  - C. Business needs
  - D. Industry best practices
- 38 What is the most secure method of configuration management?
- A. OOB
  - B. CLI
  - C. SSL
  - D. Telnet
- 39 Which of the following is not a key principle for a self-defending network?
- A. Integrated security
  - B. Collaboration
  - C. Adaptation
  - D. Firewall controls
- 40 Which of the following is not an active threat defense capability of Cisco's ASA security appliances?
- A. Security policy auditing
  - B. Application inspection and control
  - C. VPNs
  - D. Cisco Secure Desktop

## Answers

1. Two keys, a public key and a private key, exist in a key pair.
2. The same key is used to encrypt and decrypt the data with symmetric key encryption.
3. A key length of 256 bits or more is considered uncrackable.
4. MD5 outputs a 128-bit digest with variable-length input.
5. SHA outputs a 160-bit key with variable-length input.
6. MD5 is used in the digital signature process.
7. Cryptography is the study of encryption.
8. Transposition is the process of changing the order of some characters in an encryption process.
9. Data can be decrypted with the other key in the pair—in this case, the server's private key.
10. Symmetric key encryption is fast and best to use when you have large amounts of data.
11. The goal of a known-plain text attack is to discover the encryption key.
12. A cipher text-only attack attempts to crack the encryption using cryptanalysis.
13. Time and money are the two biggest concerns when attempting to break encryption using a brute-force method.
14. WinMD5 can be used to verify the integrity of a file downloaded from the Internet.
15. An x.509 certificate includes a field for Issuer and Public Key.
16. x.509 is the standard for digital certificates.
17. Different values such as 1 and 0 in an XOR process result in a value of 1.
18. CA (certificate authorities) and digital certificates are two components of a PKI.
19. Integrity ensures the data is not modified in transit.
20. A hash is a one-way encryption used to validate the integrity of a file.